SECTION 4

NORMAL PROCEDURES

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SECTION 4

NORMAL PROCEDURES

4.1 GENERAL

This section contains instructions and recommended procedures that are peculiar to the operation of this helicopter.

For definition of terms, abbreviations and symbols used in this section refer to Section 1

4.2 PREPARATION FOR FLIGHT

4.2.1 Flight Restrictions

The minimum, normal, maximum and cautionary operation ranges for the helicopter and its subsystems are indicated by instrument markings, placards and decals.

For helicopter and subsystem restrictions refer to Section 2, Limitations.

4.2.2 Flight Planning

Refer to Section 5 and 9 to determine required fuel, airspeeds and power settings for takeoff, climb, cruise, and hovering and landing data necessary to accomplish the mission.

4.2.3 Mass and Balance

The takeoff and anticipated landing gross mass and balance should be obtained before takeoff and checked against mass and load limits, and center of gravity limitations (see Section 2).

4.3 PREFLIGHT CHECK

4.3.1 General

The preflight check shall be accomplished, at the latest, before the first flight of the day and according to either the Flight Manual, the Maintenance Manual or the Pilot's Checklist.

The preflight check is not a detailed mechanical inspection, but essentially a visual check of the helicopter for correct condition.

Items marked with a star (★) need to be checked <u>each flight before takeoff</u>.

When unusual local conditions dictate, the extent and/or frequency of this check shall be increased as necessary to promote safe operation.

NOTE • The following list contains only check items for the clean configuration.

• In addition to these items, check antennas and all installed optional equipment. Make certain that relevant intermediate and special inspections, in accordance with the Maintenance Manual, are complied with.

• For optional equipment check items refer to the respective Flight Manual Supplement or Maintenance Manual Chapter 800.

4.3.2 Exterior Check

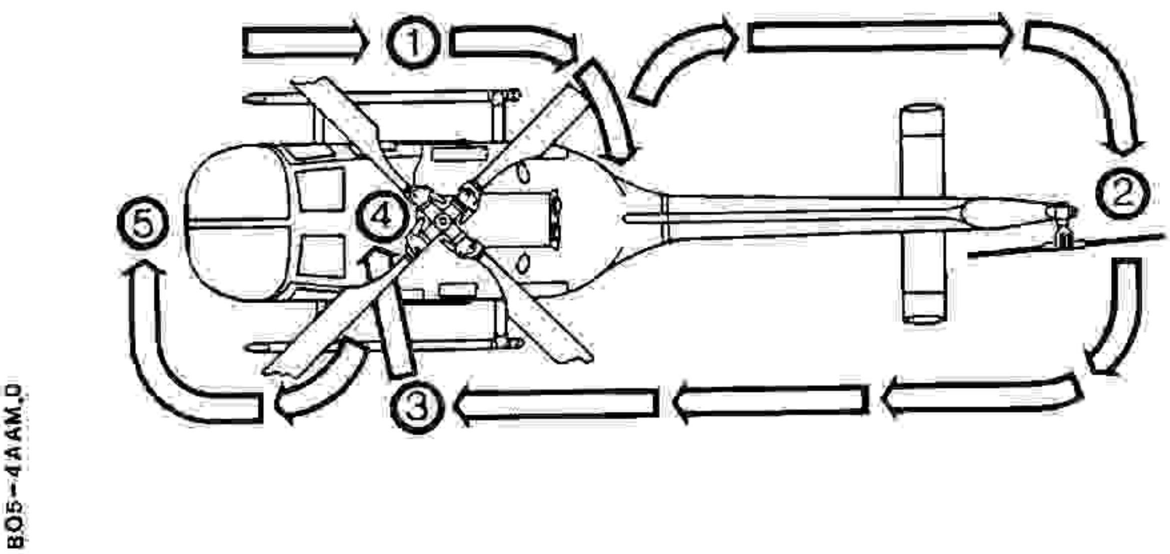
The exterior check is laid out as a walk-around check, starting forward right at the pilot's door, proceeding clockwise to the tail boom, to the left hand side (including the upper and lower areas of the helicopter) and being completed at the helicopter nose area.

NOTE • The helicopter should be headed into the wind before starting engines.

• The area around the helicopter should be clear of all foreign objects.

• To avoid excessive drain on the helicopter battery, particularly during cold weather, all ground operations should be conducted using an external power unit (EPU).

• When the battery is used, the operation of electrical equipment should be kept to a minimum.



1. FUSELAGE – RIGHT SIDE
2. TAIL BOOM
3. FUSELAGE – LEFT SIDE
4. CABIN TOP
5. NOSE

Fig. 4-1 Exterior Check Sequence

★	Helicopter forms and documents	– Check, complete
	Fuselage underside	– Condition, no fuel leaks
	Each fuel tank (3)	– Drain sample
	When OAT is above 40 °C or below –32 °C:	
	Fuel temperature	– Measure; determine fuel type to be used and/or the permissible altitude in accordance with Section 2.
★	Covers and tie-downs	– Removed
★	Ice and snow (if any)	– Removed
★	Ground handling wheels	– Removed

Cockpit

All switches	– Off
Power levers	– OFF
Rotor brake lever (if installed)	– Full down
Circuit breakers	– As required
Battery switch	– On
Voltmeter indication	– 24V DC nominal
If EPU is used:	
Battery switch	– Off
Pitot tube heating	– Function
Lighting:	
Position lights	– Function
Anti-collision lights	– Function
Landing light(s)	– Function
Before Night Flights:	
Dome and instrument lights	– Function
Utility lamp	– Function
Hand lamp	– Function
Battery switch	– Off
Hand fire extinguisher	– On board, check pressure
Flashlight	– On board, function

	First aid kit (if installed here)	– On board

Fuselage - Right Side

	Cockpit overhead window	– Condition
★	OAT indicator probe	– Condition
	Cockpit door	– Condition, function, jettison handle secured
	Pilot seat and safety belts	– Condition
★	Static port	– Clean, unobstructed
★	Pitot tube(s)	– Condition, clean, unobstructed
	Landing gear and step	– Condition
	Cabin door	– Condition, function
★	Baggage, cargo, loose items	– Stowed, secured
	Passenger seats and seat belts	– Condition
★	Transmission compartment	– No foreign objects, no leakage, free of ice/snow

EFFECTIVITY *For helicopters after SB BO105–10–126*

	Transmission oil filter clogging indicator pin	– Check in

EFFECTIVITY *ALL*

	Oil cooler	– No leakage
★	Engine oil levels	– Above ADD OIL mark
	Engine oil tanks	– Security of mounting, no leakage, filler caps secured
	Scavenge oil filter (if installed) clogging indicator pin	– Check in
★	Transmission oil level	– Above MIN mark
	Transmission	– Condition, security of mounting, no leakage, filler cap closed
	Rotor brake (if installed)	– Condition
★	Engine air intake	– Condition, no dirt, no foreign objects, free of ice/snow
	Compressor	– Rotate by hand, check for smooth rotation
★	Transmission access door	– Closed, secured

Fuselage - Right Side (Continued)

★	Engine compartment	—	No foreign objects, no fuel/oil leakage
★	Engine compartment ventilation air intake	—	Clear
	Engine	—	Condition
	Wiring, linkages and lines	—	Condition, no leakage, no chafing
	Engine mounts	—	Condition, security of mounting
	N_1-governor and connections	—	Security of mounting
	Starter/Generator	—	Security of mounting
	Engine driveshaft	—	Check freewheeling
★	Engine access door	—	Closed, secured
	Drain and vent ports	—	Clear
	Battery (if installed here)	—	Condition, no electrolyte spillage, security of mounting
	Battery connectors (2)	—	Connected
	Battery vents	—	Clear
★	Battery compartment door	—	Closed, secured
	Engine cowling	—	Secured
	Engine exhaust pipe	—	Condition, security of mounting
★	Fuselage exterior	—	Condition
	Clam shell doors and latches	—	Condition, function
	First aid kit (if installed here)	—	On board
	Avionics (in avionics equipment bay)	—	Security of mounting
★	Baggage, cargo, loose items	—	Stowed, secured
	Avionics bay door	—	Closed
★	Clam shell doors	—	Closed, secured
	Spoiler	—	Condition

Tail Boom

★	Tail boom - RH	—	Condition
★	Tail-rotor driveshaft fairing	—	Secured
	Drain ports	—	Clear

Tail Boom (Continued)

★	RH horizontal and vertical stabilizer, position light (green)	–	Condition
	Vertical fin, position light (white), anti-collision light	–	Condition
	Vertical fin cowling	–	Secured
★	Tail rotor	–	Condition
★	Pitch links	–	Condition
	PU Erosion-protective film (if fitted)	–	Condition, no separation
★	Intermediate gearbox	–	Oil level, no leakage
★	Tail-rotor gearbox	–	Condition, oil level, no leakage

NOTE Oil levels may be easier to verify by grasping tail skid and shortly shaking tail boom.

★	Intermediate gearbox access door (if installed)	–	Closed, secured
	Tail skid	–	Condition
★	LH horizontal and vertical stabilizer, position light (red)	–	Condition
★	Tail boom - LH	–	Condition
★	Tail-rotor driveshaft fairing	–	Secured
	Drain ports	–	Clear

Fuselage - Left Side

★	Fuel filler cap	–	Closed, secured
	Drain and vent ports	–	Clear
★	Fuselage exterior	–	Condition
	Engine exhaust pipe	–	Condition, security of mounting
	Engine cowling	–	Secured
★	Engine compartment	–	No foreign objects, no fuel/oil leakage
★	Engine compartment ventilation air intake	–	Clear
	Engine	–	Condition
	Wiring, linkages and lines	–	Condition, no leakage, no chafing
	Engine mounts	–	Condition, security of mounting
	N_2-governor and actuator	–	Condition, security of mounting

Fuselage - Left Side (Continued)

	Starter/Generator	–	Security of mounting
	Engine driveshaft	–	Check freewheeling
★	Engine access door	–	Closed, secured
★	Hydraulic compartment	–	No foreign objects, no leakage, free of ice/snow
★	Hydraulic module	–	No leakage
★	Area between hydraulic module and XMSN floor, especially control rod bellows	–	Free of ice/snow
	Water drain ports	–	Clear
	Hydraulic filter clogging indicator pins (4)	–	Check in
★	Hydraulic reservoir oil levels	–	Above FILL mark
★	Service-fitting caps	–	Installed
	Transmission	–	Condition, security of mounting, no leakage
★	Engine air intake	–	Condition, no dirt, no foreign objects, free of ice/snow
	Compressor	–	Rotate by hand, check for smooth rotation
★	Hydraulic compartment access door	–	Closed, secured
	Main rotor	–	Rotate by hand in direction of rotation, check for smooth rotation
★	Rotor hub oil level	–	Between MIN and MAX marks
★	Rotor blades	–	Condition
★	Rotor blade attachment bolts	–	Secured
	PU Erosion-protective film (if fitted)	–	Condition, no separation
★	Driving link assy, mixing lever assy	–	Condition, secured
★	Rotating control rods	–	Condition, free movement, secured
	Rotating control rod spherical bearings	–	Smooth operation
	Swashplate and boot (if installed)	–	Condition, secured
	Pendulum absorbers (if installed)	–	Condition, free movement
★	Air intake	–	Condition, no foreign objects, free of ice/snow

Fuselage - Left Side (Continued)

	Engine inlet deflector shield (if installed)	–	Condition, no foreign objects
	Cockpit overhead window	–	Condition
	Cabin door	–	Condition, function
★	Baggage, cargo, loose items	–	Stowed, secured
	Passenger seats and seat belts	–	Condition
	Fuel vent port	–	Clear
	Landing gear and step	–	Condition
★	Static port	–	Clean, unobstructed
	Cockpit door	–	Condition, function, jettison handle secured
	Copilot cyclic stick (if installed)	–	Secured, safety-wired
	Copilot collective lever (if installed)	–	Secured
	Copilot seat and safety belts	–	Condition
★	If copilot seat is unoccupied:		
	Copilot safety belts	–	Fastened, tightened

Nose

	Windshields	–	Condition
	Battery (if installed here)	–	Condition, no electrolyte spillage, security of mounting
	Battery connectors (2)	–	Connected
	Battery vents	–	Clear
	Nose compartment	–	No foreign objects
★	Nose compartment door	–	Closed, secured
	Cockpit air intake screens	–	Clear
	Drain ports	–	Clear
	Lower cockpit windows	–	Condition
	Fuselage bottom	–	Condition
	Anti-collision light	–	Condition
	Landing light(s)	–	Condition

4.3.3 Interior Check

★	Seat and pedals	– Adjust
★	Safety belts	– Fasten, adjust
★	Shoulder harness locks	– Function
★	Instrument panel:	
	All instruments	– Check and set
★	Switch panel:	
	Emergency fuel shutoff switches	– Guarded
	Circuit breakers	– As required
	All switches	– Off
	COMM/NAV equipment	– Off
★	Overhead panel:	
	Power levers	– OFF
	All switches	– Off
	Circuit breakers	– As required
★	Collective lever panel:	
	All switches	– Off

Pre-start Check

★	Battery switch	– On
	For EPU starts:	
	Battery switch	– Off
	EPU	– Connected
	EPU ON caution light	– On
★	Voltmeter indication .	
	Battery	– Minimum 24V
	EPU	– Between 24-28V
★	Warning/caution lights	– Test
★	Mast moment indicator	– Test
★	Fuel quantity indicator	– Check readings
★	Fuel pumps (4)	– Function
★	Collective lever lock	– Release
★	Flight controls	– Check for freedom of movement, full travel

	Hydraulic override (HY BLOCK)	– Check function

NOTE To achieve correct results, cyclic stick must be moved exactly along the longitudinal and lateral axes of the helicopter.

	Collective lever	– Raise slightly
	Hydraulic test switch	– RESET and release
	HY BLOCK warning light	– Check off
	Collective lever	– Raise until **HY BLOCK** warning light comes on, then lower slightly and release
	Hydraulic test switch	– RESET and release
	HY BLOCK warning light	– Check off
	Collective lever	– Lower until **HY BLOCK** warning light comes on, then raise slightly and release
	Hydraulic test switch	– RESET and release
	HY BLOCK warning light	– Check off
	Cyclic stick	– Move forward until **HY BLOCK** warning light comes on and release
	Hydraulic test switch	– RESET and release
	HY BLOCK warning light	– Check off
	Each remaining cyclic control direction	– Check as above
	Collective lever	– Full down
★	Collective lever lock	– Engage
★	**HY BLOCK** warning light	– Check on
	Cyclic trim system	– Check function

NOTE This check ensures that cyclic stick full travel is still guaranteed after the trim actuators have reached their full travel stop.

	Cyclic trim switch	– Press forward and hold until actuator audibly stops
	Cyclic stick	– Move fore and aft to limit stops; check for full fore/aft travel and free movement
	Each remaining cyclic control direction	– Check as above
★	Cyclic stick	– Centered
★	Rotor brake lever (if fitted)	– Full down

4.4 STARTING ENGINES

NOTE
- At ambient temperatures below +4 °C, refer to "Cold weather starting information" this section.
- Either engine may be started first.
- If, for any reason, a starting attempt is discontinued the entire starting sequence must be repeated from the beginning.
- When a starting attempt is aborted but TOT limits were not exceeded, wait 15 seconds after N_1 has returned to zero before attempting restart to permit excess fuel to drain from combustion chamber.
- If a start is not successful after three attempts, maintenance action is required.
- Before starting engines, the operational hydraulic system shall always be preselected to system 2 (collective lever lock engaged).

Fire guard (if available)	–	Posted
Rotor area	–	Clear
Anti-collision light	–	On

4.4.1 Starting First Engine

NOTE Operate only the controls of the corresponding engine.

CAUTION IMMEDIATELY ABORT START FOR ANY OF THE FOLLOWING:

- MAIN ROTOR IS NOT ROTATING AT 25% N_1. (A SECOND AND THIRD START ATTEMPT MAY BE MADE. IF POWER TURBINE STILL FAILS TO ROTATE, DISCONTINUE ATTEMPT).
- STAGNATED START (FUEL STARVATION).
- TOT RISES ABNORMALLY AND RAPIDLY APPROACHES START LIMIT (927 °C).
- TOT EXCEEDS 10-SECOND LIMIT WITHIN TEMPERATURE RANGE (811-927 °C).
- NO POSITIVE ENGINE OR TRANSMISSION OIL PRESSURE INDICATIONS UPON REACHING IDLE SPEED (N_1 59-65%).
- IDLE SPEED IS NOT REACHED WITHIN 60 SECONDS UNLESS N_1 AND N_2 ARE ACCELERATING AND TOT IS WITHIN LIMITS.
- N_2 INCREASES BEYOND N_{Ro}. A FREEWHEELING UNIT SLIPPAGE IS SUSPECTED AND MAINTENANCE ACTION IS REQUIRED.
- AUTO-ACCELERATION (BENDIX FUEL CONTROL SYSTEM INSTALLED). SEE "COLD WEATHER STARTING INFORMATION" THIS SECTION.

EFFECTIVITY *Hydraulic Systems **without** Positive Cross-over Indication*

- EITHER **HYD 1** OR **HYD 2** WARNING LIGHT REMAINS ON UPON REACHING GROUND IDLE SPEED (60% N_{Ro}).

***EFFECTIVITY** Hydraulic Systems **with** Positive Cross-over Indication*

- **HYD 2** WARNING LIGHT REMAINS ON UPON REACHING GROUND IDLE SPEED (60% N_{Ro}).

***EFFECTIVITY** ALL*

ABORT START PROCEDURE

- POWER LEVER – OFF
- FUEL PUMPS – OFF
- IGNITION SWITCH – OFF
- STARTER SWITCH – ON, AS NECESSARY TO LOWER TOT
- STARTER SWITCH – OFF

TOT – Less than 150 °C, otherwise ventilate engine

Fuel pumps (4) – On

Ignition switch – On

Starter switch – On; start clock

N_1 – Check increasing

CAUTION POWER LEVER MUST NOT BE MOVED FROM OFF POSITION UNTIL REQUIRED N_1 HAS BEEN REACHED. STARTING AT N_1 SPEEDS LESS THAN 12% INCREASES THE POSSIBILITY OF EXCEEDING ENGINE TOT LIMITS.

As N_1 accelerates through 12-15%:

If CECO fuel control installed:

Power lever – Slowly advance to a position slightly before IDLE (trim start)

If BENDIX fuel control installed:

Power lever – Move direct to a position slightly before IDLE

NOTE Even small displacements of the power lever from IDLE to OFF position may cause engine to flameout.

NOTE Do not wait for N_1 to peak out. Move power lever towards IDLE immediately upon reaching the desired N_1 speed. Delay in moving power lever may diminish battery capacity early in the start cycle.

TOT and N_1 – Monitor closely, observe TOT peak

Power lever – Lock in IDLE

N_2/N_{Ro} – Check increasing

Engine oil pressure indicators	– Check positive indication
At 58% N_1:	
Starter switch	– Off
Ignition switch	– Off
When N_2/N_{Ro} are above 40%:	
Generator switch	– On; monitor N_1
If N_1 decreases to below 59%::	
Generator	– Off
Power lever	– Advance. N_1 max 70%
Generator	– On

4.4.2 Starting Second Engine

CURRENT IND switch	– Select operating generator
Ammeter (battery start)	– Check below 100 A
Second engine	– Start following procedure for first engine
After EPU starts:	
Battery switch	– On
EPU	– Check disconnected

4.5 SYSTEM CHECKS

4.5.1★ Hydraulic Checks

WARNING
- BOTH HYDR SYSTEMS SHALL BE OPERATIONAL. TAKEOFF WITH A KNOWN OR SUSPECTED HYDR SYSTEM FAILURE IS PROHIBITED.
- TAKEOFF WITH SYSTEM 2 SELECTED AS THE OPERATING SYSTEM IS PROHIBITED (THERE IS NO AUTOMATIC CROSS-OVER FROM SYSTEM 2 TO SYSTEM 1).
- DO NOT OPERATE THE HYDRAULIC TEST SWITCH DURING FLIGHT.

NOTE
- Hydraulic system checks may be performed between starting first and second engine at pilot's discretion.
- System 2 is always the operating hydraulic system after starting engines because the collective lever lock is engaged (collective-down micro switch activated).

***EFFECTIVITY** Hydraulic Systems **without** Positive Cross-over Indication*

HYD 1 and **HYD 2** warning lights	–	Off
HY BLOCK warning light	–	Check on

***EFFECTIVITY** Hydraulic Systems **with** Positive Cross-over Indication*

HYD 2 warning light	–	Off
HYD 1 and **HY BLOCK** warning lights	–	Check on

***EFFECTIVITY** ALL*

System 2 Check

	–	
Cyclic stick	–	Check ease of movement (not more than 3 cm from neutral) all four directions, return to neutral
Collective lever lock	–	Release
Collective lever	–	Check ease of movement, return to full down

Cross-over Check

Hydraulic test switch	–	Place in RESET, then release

***EFFECTIVITY** Hydraulic Systems **without** Positive Cross-over Indication*

HY BLOCK warning light	–	Off
HYD 1 warning light	–	Monitor closely; light must flash

NOTE
- If flashing of the **HYD 1** warning light is not noticed, place the hydraulic test switch in HYDR 2 position and repeat the system check making a short-duration cyclic stick displacement either prior to or during the cross-over.
- If the **HYD 1** warning light does not flash then repeat the cross-over check while a qualified person visually monitors the hydraulic system to verify that control has returned to System 1.
- Helicopters having optional hydraulic-powered equipment or systems, the **HYD 2** warning light may also come on for approx 2 seconds.

***EFFECTIVITY** Hydraulic Systems **with** Positive Cross-over Indication*

HYD 1 and **HY BLOCK** warning lights	–	Off

NOTE Helicopters having optional hydraulic-powered equipment or systems, the **HYD 2** warning light may also come on for approx 2 seconds.

***EFFECTIVITY** ALL*

Hydraulic test switch guard	–	Close

System 1 Check

Cyclic stick	–	Check ease of movement all four directions, return to neutral
Collective lever	–	Check ease of movement, return to full down and lock
HYD 1, **HYD 2** and **HY BLOCK** warning lights	–	Check off

4.5.2 Electrical and Avionic Checks

Electrical system	–	Functional check
Each generator in turn	–	Off
Voltmeter	–	Check not less than 27 V each generator
Respective **GEN** warning light	–	On
Both generators	–	On, warning lights off
CURRENT IND switch	–	Select each GEN position and check that the loads are approx equal, leave switch in BUS BAR (BAT) position
★ COMM/NAV equipment	–	On and check
★ All other instruments and equipment	–	Check and set

4.5.2 Miscellaneous Checks

ANTI-ICING system	–	Check as required
ENG. 1/ENG. 2 switches	–	On
ANTI ICING 1/ANTI ICING 2 or **L/R** caution lights	–	On
TOT	–	Note increase
ENG. 1/ENG. 2 switches	–	Off
ANTI ICING 1/ANTI ICING 2 or **L/R** caution lights	–	Off
TOT	–	Note decrease

4.6 ★ PRE-TAKEOFF CHECK

4.6.1★ On Ground

Crew and passengers	–	Briefed
Passenger safety belts	–	Fastened
Doors	–	Closed, locked
Warning/caution lights	–	Off

NOTE **OIL COOL** warning light (if installed) may remain on until after placing power levers full forward.

Cyclic stick	–	Check centering device secured

CAUTION IF ENGINES HAVE BEEN SHUT DOWN FOR MORE THAN 15 MINUTES, STABILIZE AT IDLE SPEED FOR 1 MINUTE.

Power levers	–	Smoothly to FLIGHT
OIL COOL warning light (if installed)	–	Check off
N_R/N_2	–	Check RPM warning and low rotor RPM audio tone on between 75% and 95%
	–	Trim to 98%

NOTE Adjust the volume of the beeping tone as required..

Torque	–	Synchronize
Engine and XMSN instruments	–	Check in green range
Fuel quantity indicator	–	Check readings
Voltmeter	–	Check 27–29V
Ammeter	–	Check Δ I (LH–RH) < 10 A
Bleed air consumers	–	As required
Optional equipment controls	–	As required
Collective lever lock	–	Check released
Collective lever friction	–	Adjust as required

4.6.2★ At Hover

Torque	–	Synchronize
Hover power	–	Check
N_2/N_{Ro}	–	Adjust to 98-102%

NOTE 100% rotor RPM recommended for steady flight condition.

Engine and XMSN instruments	–	Check in green range
All warning/caution lights	–	Off

4.7 CRUISE CHECK

NOTE Recommended maximum values for use under normal cruise conditions are 82% Torque and 738 °C TOT.

Power check	–	Perform as required or when deemed necessary (see Section 5).

4.8 PRE-LANDING CHECK

N_2/N_{Ro}	–	Adjust to 98-102%

NOTE 100% rotor RPM recommended during landing.

All instruments	–	Check
Bleed air consumers	–	As required
All warning/caution lights	–	Check

4.9 ENGINE SHUTDOWN

4.9.1 Normal

NOTE After first flight of the day perform **Deceleration Check**.

Collective lever lock	–	Engage
Power levers	–	IDLE; start clock
Cyclic stick	–	Trim neutral
	–	
All electrical consumers	–	Off except anti-collision lights and fuel pumps
After 2-min IDLE:		
Power levers	–	OFF

CAUTION IF TOT DOES NOT DECREASE WITH N_1, CLOSE EMERG FUEL VALVE AND, WHILE HOLDING POWER LEVER OFF, VENTILATE AFFECTED ENGINE UNTIL TOT SUBSTANTIALLY DECREASES. MAINTENANCE ACTION IS REQUIRED BEFORE NEXT START.

TOT and N_1	–	Monitor decreasing
Fuel pumps (4)	–	Off
Generator switches	–	Off
OIL COOL warning light (if installed)	–	Check on below 20% N_{Ro}
Circuit breakers	–	As required
All switches	–	Off

4.9.2 **Deceleration Check**

NOTE Perform this check after first flight of each day and for each engine separately.

Cyclic stick	–	Trim neutral
Collective lever lock	–	Engage
All electrical consumers	–	Off except anti-collision lights and fuel pumps
Generator switches	–	Off
N_2 (BENDIX fuel controls only)	–	Stabilize at 100% for 15 seconds
Power lever of check engine	–	Snap to IDLE, start clock (stop watch)

When N_1 of check engine passes 65%:

Clock	–	Stop, note time

NOTE
- Minimum allowable deceleration time is 2 seconds. If deceleration time is less than 2 seconds, perform two more checks to confirm time.
- Maintenance action is required when:
  - confirmed deceleration time is less than 2 sec,
  - N_1 drops below 59%,
  - flameout is experienced.

Power lever of check engine	–	FLIGHT
Remaining engine	–	Perform deceleration check
Both power levers	–	IDLE
	–	

After 2 min:

Power levers	–	OFF

CAUTION IF TOT DOES NOT DECREASE WITH N_1, CLOSE EMERG FUEL VALVE AND, WHILE HOLDING POWER LEVER OFF, VENTILATE AFFECTED ENGINE UNTIL TOT SUBSTANTIALLY DECREASES. MAINTENANCE ACTION IS REQUIRED BEFORE NEXT START.

TOT and N1	–	Monitor decreasing
Fuel pumps (4)	–	Off
OIL COOL warning light (if installed)	–	Check on below 20% N_{Ro}
Circuit breakers	–	As required
All switches	–	Off
Engine compressor rinse	–	Perform as required

4.10 COLD WEATHER STARTING INFORMATION

Following are some general practices recommended for improved cold weather starting:

EFFECTIVITY *CECO fuel control system*

If engines equipped with CECO fuel control system start and then flame out at altitudes above 5000-ft MSL during cold ambient conditions, attempt another start with fuel supply pumps off. After reaching idle RPM, switch fuel supply pumps on.

EFFECTIVITY *BENDIX fuel control system*

CAUTION CONDENSATION AND FREEZING OF MOISTURE IN THE FUEL CONTROL PNEUMATIC CIRCUITS CAN OCCUR WHEN WEATHER CONDITIONS OF LOW TEMPERATURE AND HIGH RELATIVE HUMIDITY ARE ENCOUNTERED. THIS CONDITION CAN CAUSE AUTO (SPONTANEOUS) ACCELERATION OF THE AFFECTED ENGINE.

Auto acceleration resulting from frozen moisture in the fuel system air circuits does not repeat unless the aircraft is again subjected to cold soak. To prevent icing/auto acceleration, make a 10-minute ground warm-up run at IDLE before flight. This warm-up is recommended when the aircraft has been allowed to cold soak (remain out of hangar overnight) in low ambient temperature of –12 °C or below and high relative humidity 45% or greater.

If auto acceleration occurs, perform the abort start procedures. Subsequently, restart the engine and resume the warm-up.

EFFECTIVITY *ALL*

When engine and main-rotor transmission oil temperatures are below –25 °C and OAT is below –30 °C preheat the oils to a temperature of –25 °C or higher.

If available and conditions allow when the aircraft has been cold soaked at temperatures below 4 °C, use an EPU to assist for faster more satisfactory starts.

If the aircraft has been cold soaked at temperatures below −18 °C and a battery start must be made, preheat the engine fuel control area and battery if equipment is available and conditions allow.

If the aircraft has been cold soaked and a battery start must be made without preheating the battery, remove and store the battery until it is required if conditions allow. Store the battery in an area where it can be maintained or warmed to a temperature above ambient outside conditions or to approx 20 °C.

If stagnated starts are encountered, maintenance action is required.

Due to variations in jet fuels available for commercial operation during cold weather starting, the engines may experience a short delay before ignition after each power lever is moved to just before the IDLE position. This delay should be less than 3 seconds regardless of the type of fuel used. If the ignition delay exceeds 3 seconds, return the power lever to OFF and ventilate the engine for 30 seconds to remove excessive fuel from the combustion section.

In some instances N_1 may accelerate slowly through the 25-30% speed range on a battery start after an engine has been cold soaked and not preheated. If the start is not completed within the starter engagement time limits, shut down the engine. Before attempting the next start, wait for at least one minute to pass. This wait will allow residual heat from the previous start attempt to soak back into the engine and battery and improve conditions for the next start attempt.

At temperatures below −30 °C OAT, initial oil pressure indication may be delayed for up to 10 seconds on each engine. The final pressure build-up indication of the LH engine may be delayed for up to 50 seconds.

Depending upon the OAT, a ground idle time of up to 20 minutes may be necessary to bring engine and transmission oils up to operating temperatures.

4.11 FLIGHT CHARACTERISTICS

4.11.1 Flight Controls

CAUTION AVOID EXTREME CYCLIC STICK DISPLACEMENTS WHEN ON THE GROUND WITH ROTOR TURNING.

During ground operations with rotor turning, the cyclic stick must remain in the neutral position and the collective pitch lever in the full down position; however, for functional test purposes, minimum control movements (not more than 3 cm from neutral) are allowed.

Avoid extreme pedal movements during ground operations.

4.11.2 Lateral Control Characteristics

WARNING AVOID STEEP RIGHT TURNS AT AIRSPEEDS BELOW 70 KIAS CLOSE TO THE GROUND IN ORDER TO MAINTAIN SUFFICIENT LATERAL CONTROL MARGIN FOR RECOVERY.

Lateral control characteristics of single-rotor helicopters are significantly influenced by forward speed (airspeed), bank angles and relating g-loads.

During high load factor maneuvers at airspeeds below 70 KIAS the cyclic stick can reach the left lateral stop before retreating-blade stall entry.

Reduction of the load factor and/or nose-left side slip will improve left lateral control margin.

The recommended maximum bank angles for left and right turns at various airspeeds are as follows:

AIRSPEED (KIAS)	MAX ANGLE OF BANK
0 - 30	30°
30 - 60	45°
60 - (V_{NE}* – 30)	60°
(V_{NE}* – 30) - V_{NE}*	30°

* For V_{NE} values, refer to the V_{NE} table in Section 2.